Updated Project Definition

High Level Description

The Personalized Weather Station is a compact embedded environmental monitoring system designed to collect real-time weather data and send it to a user-facing computer application. The system measures temperature, humidity, and atmospheric pressure using a BME280 sensor connected to an ATmega328P-A microcontroller. The board communicates with a computer over USB serial communication, where a Python program receives the data, graphs the readings, estimates short-term trends, classifies the weather condition, and can send phone notifications through ntfy.

The project also serves as a full PCB design exercise. It includes schematic design, PCB routing, BOM generation, manufacturing file preparation, debugging, and documentation. The design is intended to be manufacturable and expandable for future features such as improved prediction, app integration, wireless communication, cloud logging, and enclosure development.

Purpose

The purpose of the project is to design a smaller and more personalized alternative to oversized weather monitoring systems while also gaining practical experience with embedded hardware and PCB manufacturing.

The system is meant to:

- Monitor environmental conditions in real time
- Measure temperature, humidity, and atmospheric pressure
- Use a custom PCB instead of a full development-board-only setup
- Practice schematic capture, PCB layout, ERC/DRC checks, BOM creation, and manufacturing preparation
- Use USB serial communication so a computer program can graph and process the sensor data
- Explore prediction logic and phone notifications as future-facing product features

Objectives

- Design a custom PCB for a personalized weather station
- Use an ATmega328P-A microcontroller as the main embedded controller
- Use a BME280 sensor for temperature, humidity, and pressure measurement
- Add the required support circuitry including reset pull-up, pull-up resistors, crystal oscillator, decoupling capacitors, voltage regulation, USB-C connection, and ISP programming header
- Create a BOM and prepare the board for manufacturing

- Validate the concept through testing and demonstration software
- Use Python to receive USB serial data, graph actual readings, estimate predicted values, classify weather trends, and send ntfy notifications
- Document design choices, troubleshooting steps, and lessons learned

System Overview

The system is built around the ATmega328P-A microcontroller. The BME280 sensor collects environmental data and sends it to the microcontroller. The board includes supporting components for stable operation including:

- External crystal oscillator
- Load capacitors
- Reset circuitry
- Decoupling capacitors
- Test points
- ISP programming header
- USB-C hardware
- CH340C USB-to-serial interface
- 3.3 V regulator

The PCB currently includes two dedicated USB-C connectors:

- One USB-C connector for board power input
- One USB-C connector for communication and USB serial data transfer

The computer-side Python program connects to the board using USB serial communication. Once connected, it reads temperature, humidity, and pressure values, converts the units, graphs the readings, predicts short-term trends, classifies the weather condition, and sends phone notifications through ntfy during the demo.

Major Components

Component	Purpose
ATmega328P-A	Main microcontroller
BME280	Temperature, humidity, and pressure sensing
CH340C	USB-to-serial communication
USB-C Connectors	Power and communication interfaces
LD1117S33TR	3.3V voltage regulator
Crystal Oscillator	Stable MCU timing
ISP Header	Firmware programming

Test Points	Validation and debugging
LED + Resistor	Status indication

Actual Development Path

The project did not stay exactly the same as the original proposal. The original concept mentioned a Raspberry Pi 4, DHT22, GUI display, and future BME280 pressure integration. What actually happened was that the DHT22 was used only as an early proof of concept, and the project moved to the BME280 because it measures temperature, humidity, and pressure in one sensor.

The Raspberry Pi/touchscreen-style product idea was also moved away from because it was too large and expensive for a realistic weather station product.

The final direction became a custom PCB with an ATmega328P-A and BME280 connected to a computer by USB serial communication. The computer application handles the higher-level user interface, graphing, prediction, and phone notifications.

Testing and Demonstration

Testing focused on proving that the sensing and data processing concept worked and that the PCB design was manufacturable.

The Python program:

- Received serial data from the device
- Displayed a Tkinter interface
- Graphed live environmental data
- Predicted short-term trends
- Classified weather behavior
- Sent ntfy notifications

The demonstration showed how a user could:

- Plug the device into a computer
- Run the application
- View environmental readings
- Monitor predictions and notifications automatically

Troubleshooting and Design Challenges

- The original proof of concept used a DHT22 because it was available and easy to start with
- The DHT22 was later replaced with the BME280
- Some BME280 modules were damaged during testing because the pins were incorrectly labeled or misidentified

- The initial PCB layout required review because support components were missing
- Test points were added for easier validation and debugging
- USB-C communication was selected instead of Wi-Fi hardware because wireless communication was unnecessary for the current demo
- A CH340C USB-to-serial interface was added to support communication with the computer application
- A 3.3 V regulator was included because the sensor and MCU required stable lower-voltage operation
- The project explored 4-layer PCB routing concepts to better understand professional PCB workflows
- The final direction moved away from the Raspberry Pi and touchscreen concept because it was too bulky and costly

Future Improvements

- Fabricate and assemble the PCB, then validate it with real measurements
- Create a cleaner desktop or mobile application
- Improve prediction logic using a better regression model
- Add cloud logging or data export
- Design a custom enclosure
- Add waterproofing or outdoor sensor protection
- Improve power management for battery operation
- Explore wireless communication in future revisions if needed

The Personalized Weather Station successfully evolved from a simple proof-of-concept idea into a more realistic custom PCB-based embedded system. The final direction used an ATmega328P-A, BME280 sensor, USB serial communication, Python graphing, prediction logic, and phone notifications.

The project provided hands-on experience with:

- PCB design
- Embedded system development
- Hardware troubleshooting
- Manufacturing preparation
- USB communication
- Engineering documentation